LOGISTICS DELIVERY PERFORMANCE ANALYSIS

SQL-Based Exploratory & Performance Analysis

25,000 Logistics Delivery Records | PostgreSQL | Data Analytics

Project Title

Logistics Delivery Performance Analysis

Business Problem Statement

The objective of this project is to analyze 25,000 logistics delivery records to evaluate operational performance, identify factors contributing to delivery delays and failed deliveries, compare delivery partners, assess the impact of weather conditions and distance on delivery performance, analyze delivery costs, and understand customer satisfaction through delivery ratings.

This analysis helps logistics companies:

- Improve on-time delivery performance
- Reduce delivery delays and failures
- Optimize delivery costs
- Identify top-performing delivery partners
- Increase customer satisfaction
- Make data-driven business decisions

Dataset Overview

Total Records	25,000
Total Columns	15
Primary Key	delivery_id
Dataset Format	CSV

Database	PostgreSQL
Domain	Logistics and Supply Chain
Analysis Type	Exploratory and Performance Analysis

Columns Included

- delivery_id
- delivery_partner
- package_type
- vehicle_type
- delivery_mode
- region
- weather_condition
- distance_km
- package_weight_kg
- delivery_time_hours
- expected_time_hours
- delayed
- delivery_status
- delivery_rating
- delivery_cost

Tools Used

- PostgreSQL
- pgAdmin 4
- Microsoft Excel / CSV
- Microsoft Word
- GitHub
- LinkedIn

SQL Skills Used

- SELECT
- WHERE

- DISTINCT
- COUNT()
- COUNT(DISTINCT)
- AVG()
- SUM()
- MIN()
- MAX()
- ROUND()
- CASE WHEN
- GROUP BY
- ORDER BY
- LIMIT
- Column Aliases (AS)
- Conditional Aggregation
- Percentage Calculations
- Window Functions (RANK())
- Data Categorization
- Business KPI Calculations

KPIs Analyzed

- Total Deliveries
- Delay Percentage
- Failure Rate
- Delivery Status Distribution
- Average Delivery Time
- Average Delivery Cost
- Average Delivery Rating
- Cost per Kilometer
- Cost per Kilogram
- Delay Percentage by Weather Condition
- Delay Percentage by Distance Category

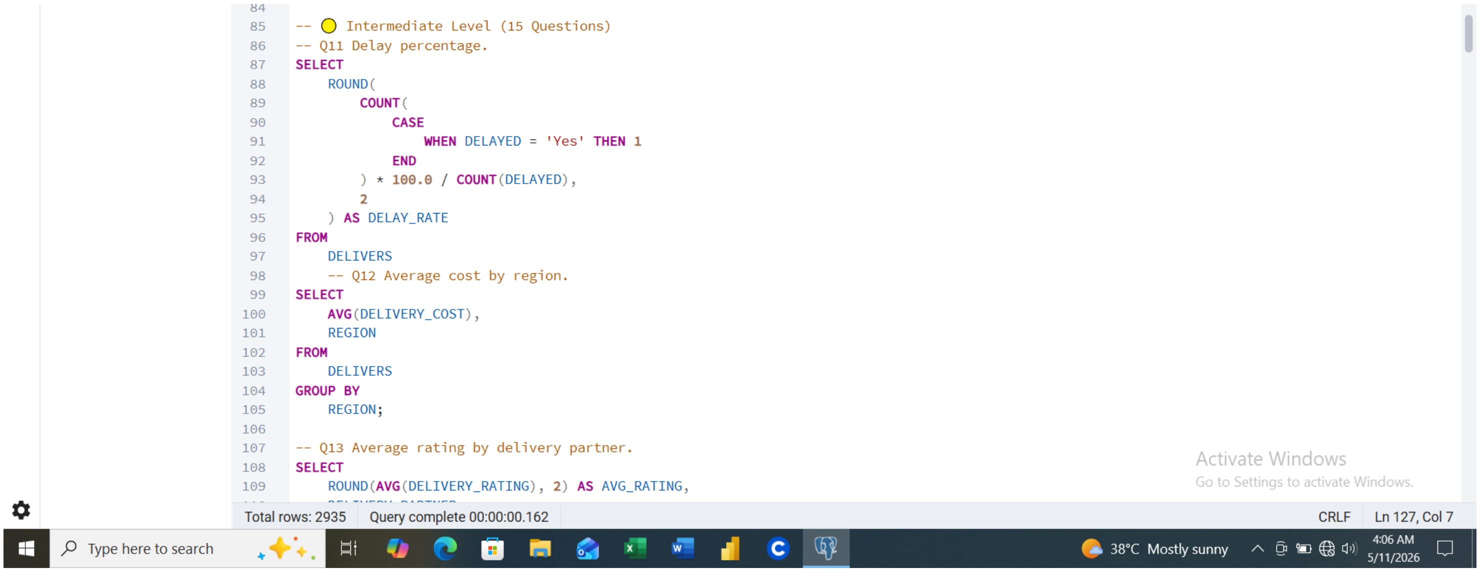
SQL Analysis Screenshots

Basic SQL Queries (Q1–Q10)

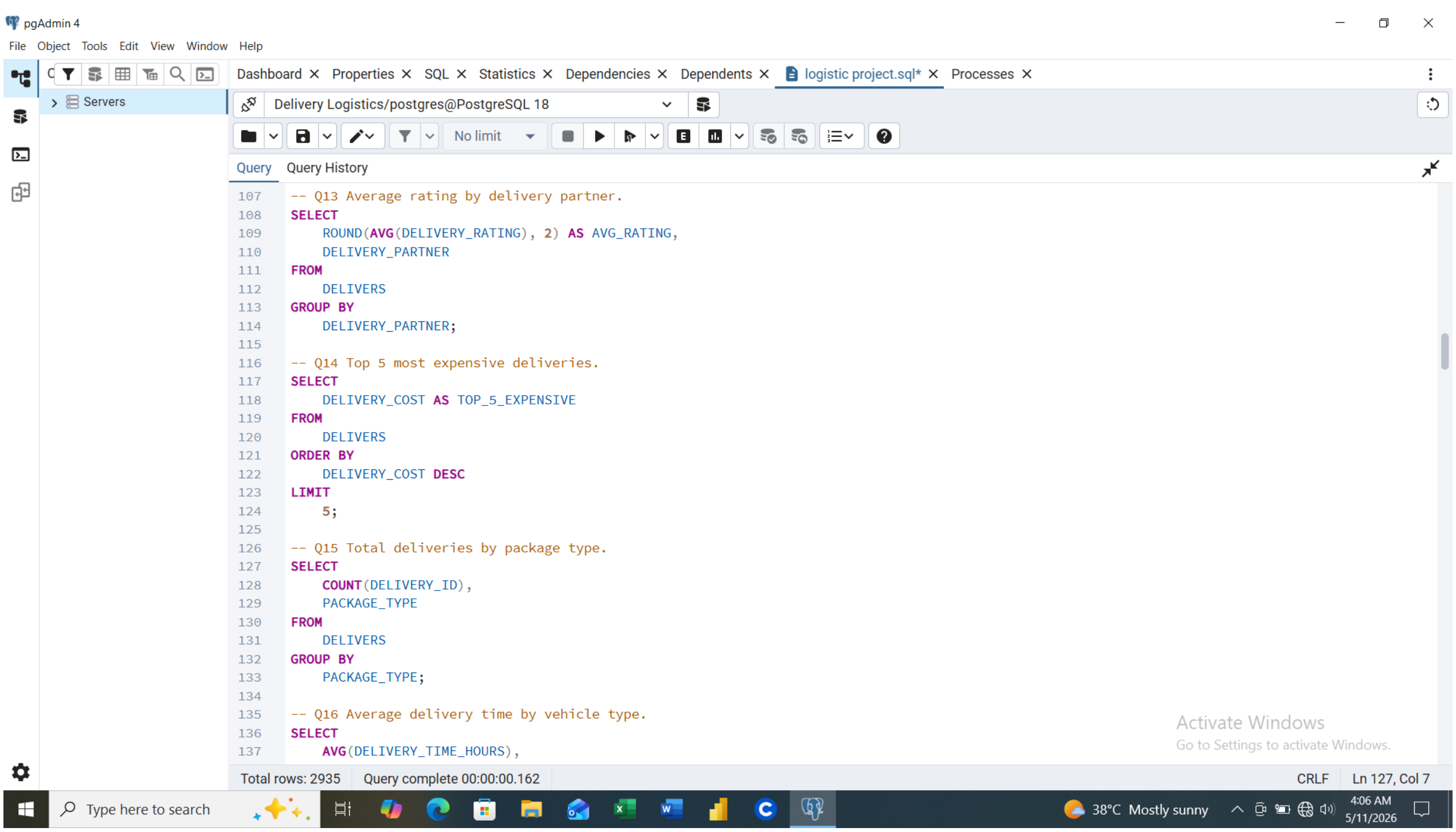
```sql
-- 🟢 Beginner Level (10 Questions)
-- Q1 Count total deliveries.
SELECT
    COUNT(*)
FROM
    DELIVERS;

-- Q2 Count unique delivery partners.
SELECT
    COUNT(DISTINCT DELIVERY_PARTNER)
FROM
    DELIVERS;

-- Q3 Show all distinct regions.
SELECT DISTINCT
    REGION
FROM
    DELIVERS;

-- Q4 Average delivery rating.
SELECT
    AVG(DELIVERY_RATING)
FROM
    DELIVERS;

-- Q5 Maximum delivery cost.
SELECT
    MAX(DELIVERY_COST) AS MAX_COST
FROM
    DELIVERS;

-- Q6 Minimum delivery time.
SELECT
    MIN(DELIVERY_TIME_HOURS)
FROM
    DELIVERS;

-- Q7 Count delayed deliveries.
SELECT
    COUNT(DELAYED) AS DELAYED_YES
FROM
    DELIVERS
WHERE
    DELAYED = 'Yes';

-- Q8 Count failed deliveries.
SELECT
    COUNT(*) AS FAILED_DELIVERYS
FROM
    DELIVERS
WHERE
    DELIVERY_STATUS = 'Failed';

-- Q9 Average package weight.
SELECT
    AVG(PACKAGE_WEIGHT_KG)
FROM
    DELIVERS;

-- Q10 Average distance.
SELECT
    AVG(DISTANCE_KM)
FROM
    DELIVERS;
```

Total rows: 2935 Query complete 00:00:00.162

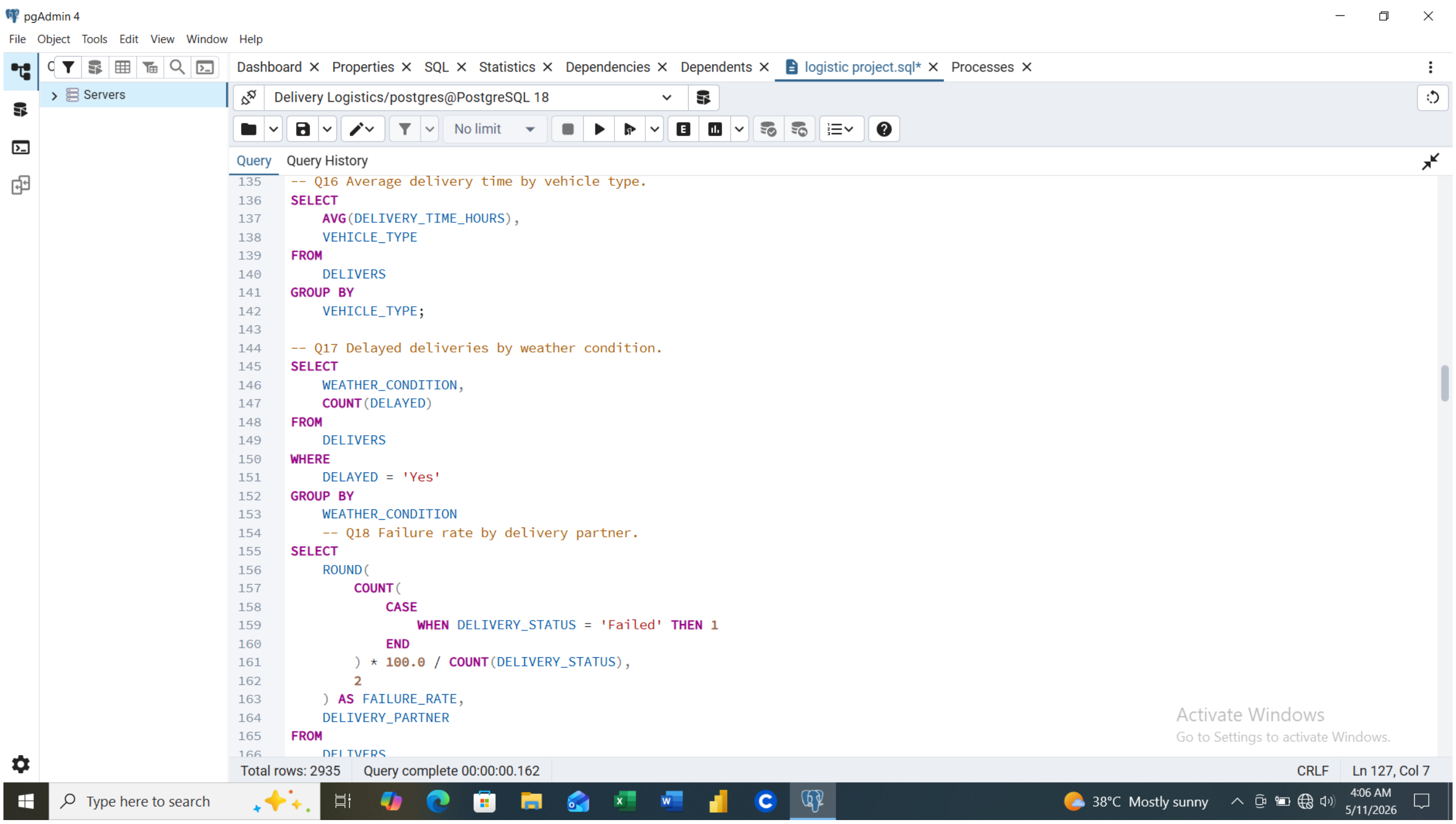
Intermediate SQL Queries (Q11–Q25)



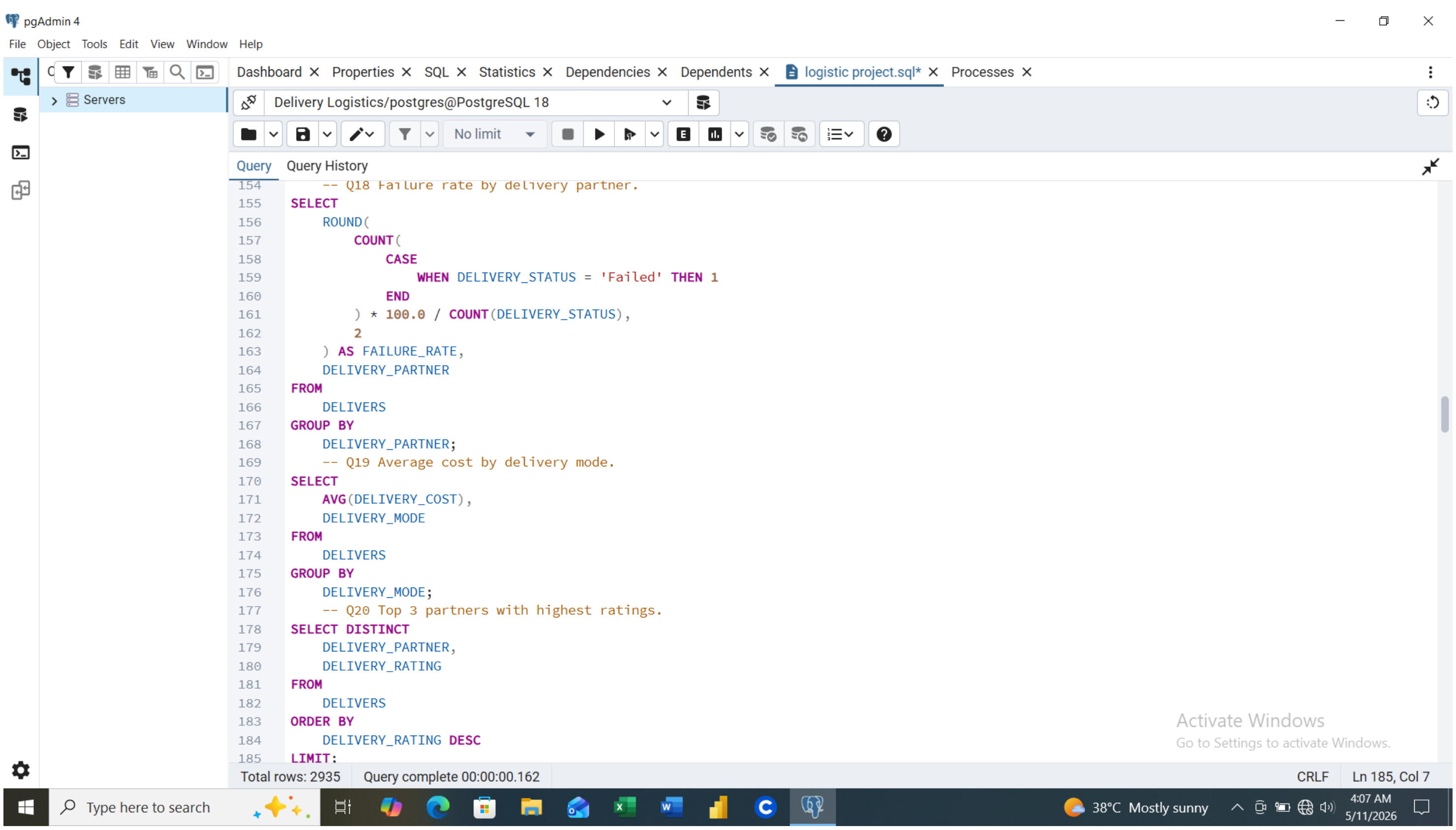
```sql
-- 🟡 Intermediate Level (15 Questions)
-- Q11 Delay percentage.
SELECT
    ROUND(
        COUNT(
            CASE
                WHEN DELAYED = 'Yes' THEN 1
            END
        ) * 100.0 / COUNT(DELAYED),
        2
    ) AS DELAY_RATE
FROM
    DELIVERS
    -- Q12 Average cost by region.
SELECT
    AVG(DELIVERY_COST),
    REGION
FROM
    DELIVERS
GROUP BY
    REGION;

-- Q13 Average rating by delivery partner.
SELECT
    ROUND(AVG(DELIVERY_RATING), 2) AS AVG_RATING,
```

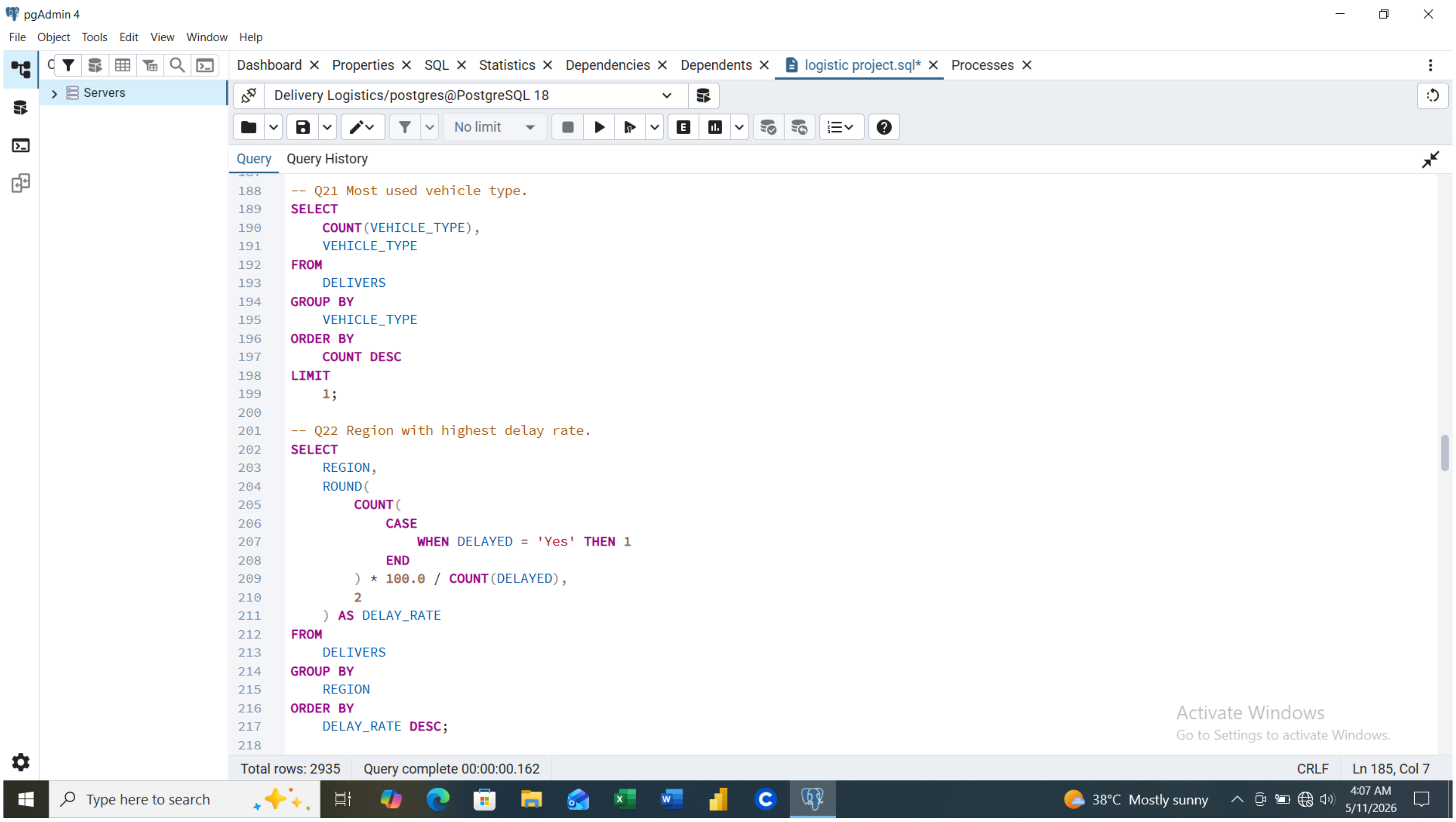



```sql
-- Q13 Average rating by delivery partner.
SELECT
    ROUND(AVG(DELIVERY_RATING), 2) AS AVG_RATING,
    DELIVERY_PARTNER
FROM
    DELIVERS
GROUP BY
    DELIVERY_PARTNER;

-- Q14 Top 5 most expensive deliveries.
SELECT
    DELIVERY_COST AS TOP_5_EXPENSIVE
FROM
    DELIVERS
ORDER BY
    DELIVERY_COST DESC
LIMIT
    5;

-- Q15 Total deliveries by package type.
SELECT
    COUNT(DELIVERY_ID),
    PACKAGE_TYPE
FROM
    DELIVERS
GROUP BY
    PACKAGE_TYPE;

-- Q16 Average delivery time by vehicle type.
SELECT
    AVG(DELIVERY_TIME_HOURS),
```

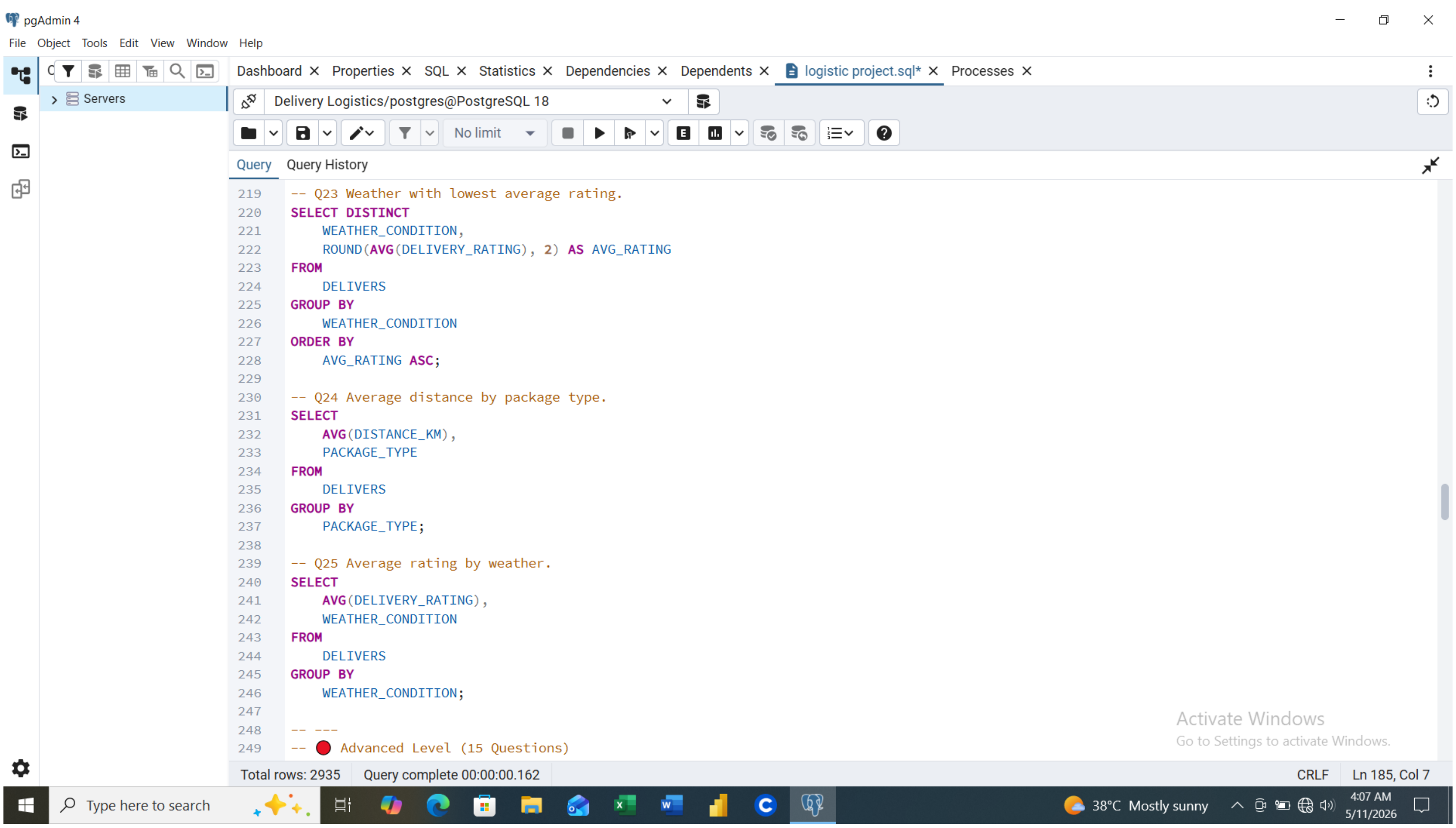

pgAdmin 4
File Object Tools Edit View Window Help
Dashboard Properties SQL Statistics Dependencies Dependents logistic project.sql* Processes
Servers
Delivery Logistics/postgres@PostgreSQL 18
No limit
Query Query History
-- Q16 Average delivery time by vehicle type.
SELECT
    AVG(DELIVERY_TIME_HOURS),
    VEHICLE_TYPE
FROM
    DELIVERS
GROUP BY
    VEHICLE_TYPE;

-- Q17 Delayed deliveries by weather condition.
SELECT
    WEATHER_CONDITION,
    COUNT(DELAYED)
FROM
    DELIVERS
WHERE
    DELAYED = 'Yes'
GROUP BY
    WEATHER_CONDITION
    -- Q18 Failure rate by delivery partner.
SELECT
    ROUND(
        COUNT(
            CASE
                WHEN DELIVERY_STATUS = 'Failed' THEN 1
            END
) * 100.0 / COUNT(DELIVERY_STATUS),
        2
) AS FAILURE_RATE,
    DELIVERY_PARTNER
FROM
Activate Windows
Go to Settings to activate Windows.
Total rows: 2935 Query complete 00:00:00.162
CRLF Ln 127, Col 7
Type here to search
38°C Mostly sunny
4:06 AM
5/11/2026


pgAdmin 4
File Object Tools Edit View Window Help
Dashboard Properties SQL Statistics Dependencies Dependents logistic project.sql* Processes
Servers
Delivery Logistics/postgres@PostgreSQL 18
No limit
Query Query History
    -- Q18 Failure rate by delivery partner.
SELECT
    ROUND(
        COUNT(
            CASE
                WHEN DELIVERY_STATUS = 'Failed' THEN 1
            END
) * 100.0 / COUNT(DELIVERY_STATUS),
        2
) AS FAILURE_RATE,
    DELIVERY_PARTNER
FROM
    DELIVERS
GROUP BY
    DELIVERY_PARTNER;
    -- Q19 Average cost by delivery mode.
SELECT
    AVG(DELIVERY_COST),
    DELIVERY_MODE
FROM
    DELIVERS
GROUP BY
    DELIVERY_MODE;
    -- Q20 Top 3 partners with highest ratings.
SELECT DISTINCT
    DELIVERY_PARTNER,
    DELIVERY_RATING
FROM
    DELIVERS
ORDER BY
    DELIVERY_RATING DESC
LIMIT:
Activate Windows
Go to Settings to activate Windows.
Total rows: 2935 Query complete 00:00:00.162
CRLF Ln 185, Col 7
Type here to search
38°C Mostly sunny
4:07 AM
5/11/2026

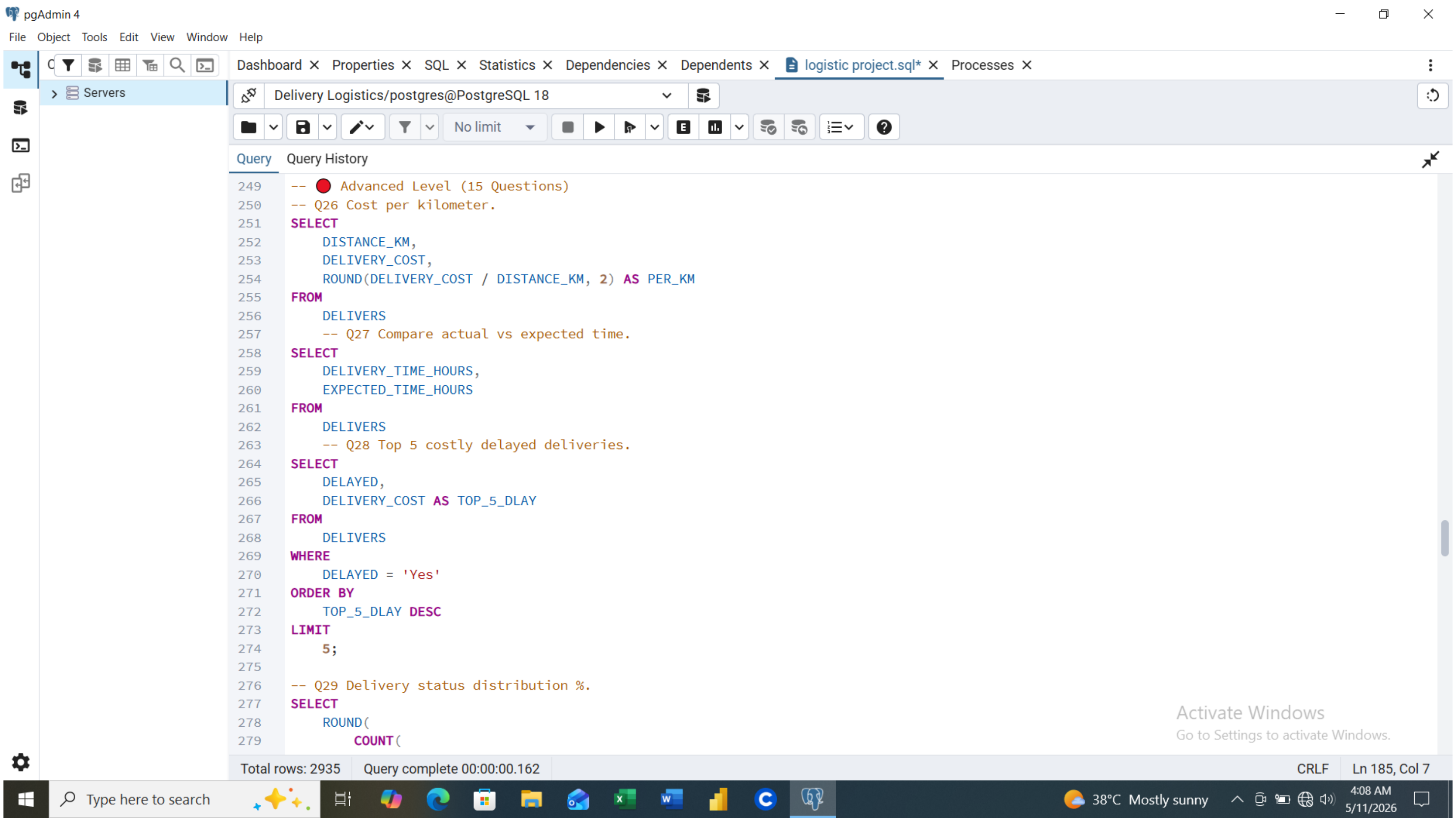


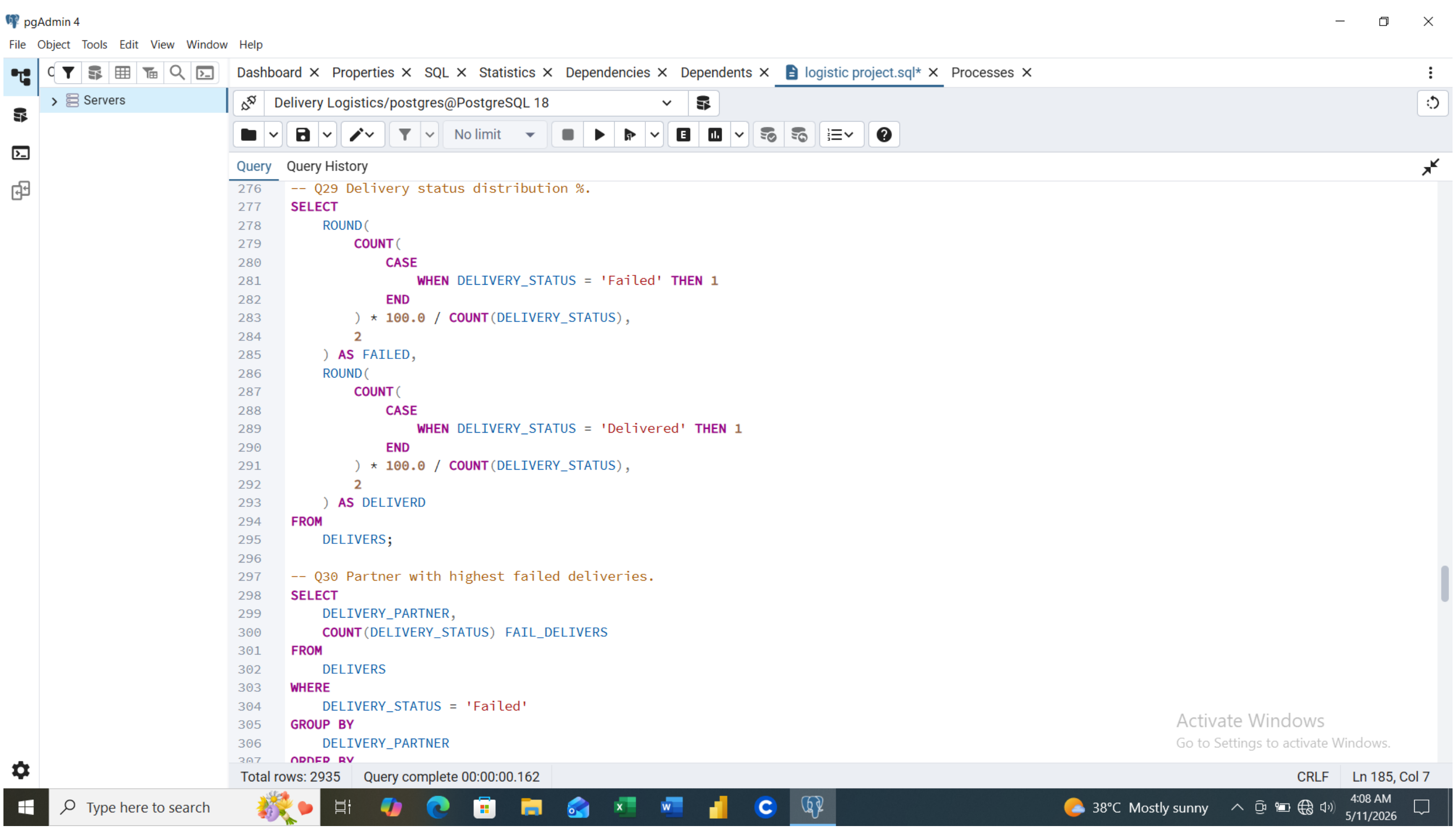

```sql
-- Q21 Most used vehicle type.
SELECT
    COUNT(VEHICLE_TYPE),
    VEHICLE_TYPE
FROM
    DELIVERS
GROUP BY
    VEHICLE_TYPE
ORDER BY
    COUNT DESC
LIMIT
    1;

-- Q22 Region with highest delay rate.
SELECT
    REGION,
    ROUND(
        COUNT(
            CASE
                WHEN DELAYED = 'Yes' THEN 1
            END
        ) * 100.0 / COUNT(DELAYED),
        2
    ) AS DELAY_RATE
FROM
    DELIVERS
GROUP BY
    REGION
ORDER BY
    DELAY_RATE DESC;
```




```sql
-- Q23 Weather with lowest average rating.
SELECT DISTINCT
    WEATHER_CONDITION,
    ROUND(AVG(DELIVERY_RATING), 2) AS AVG_RATING
FROM
    DELIVERS
GROUP BY
    WEATHER_CONDITION
ORDER BY
    AVG_RATING ASC;

-- Q24 Average distance by package type.
SELECT
    AVG(DISTANCE_KM),
    PACKAGE_TYPE
FROM
    DELIVERS
GROUP BY
    PACKAGE_TYPE;

-- Q25 Average rating by weather.
SELECT
    AVG(DELIVERY_RATING),
    WEATHER_CONDITION
FROM
    DELIVERS
GROUP BY
    WEATHER_CONDITION;

-- ---
-- 🔴 Advanced Level (15 Questions)
```

Advanced SQL Queries (Q26–Q38)



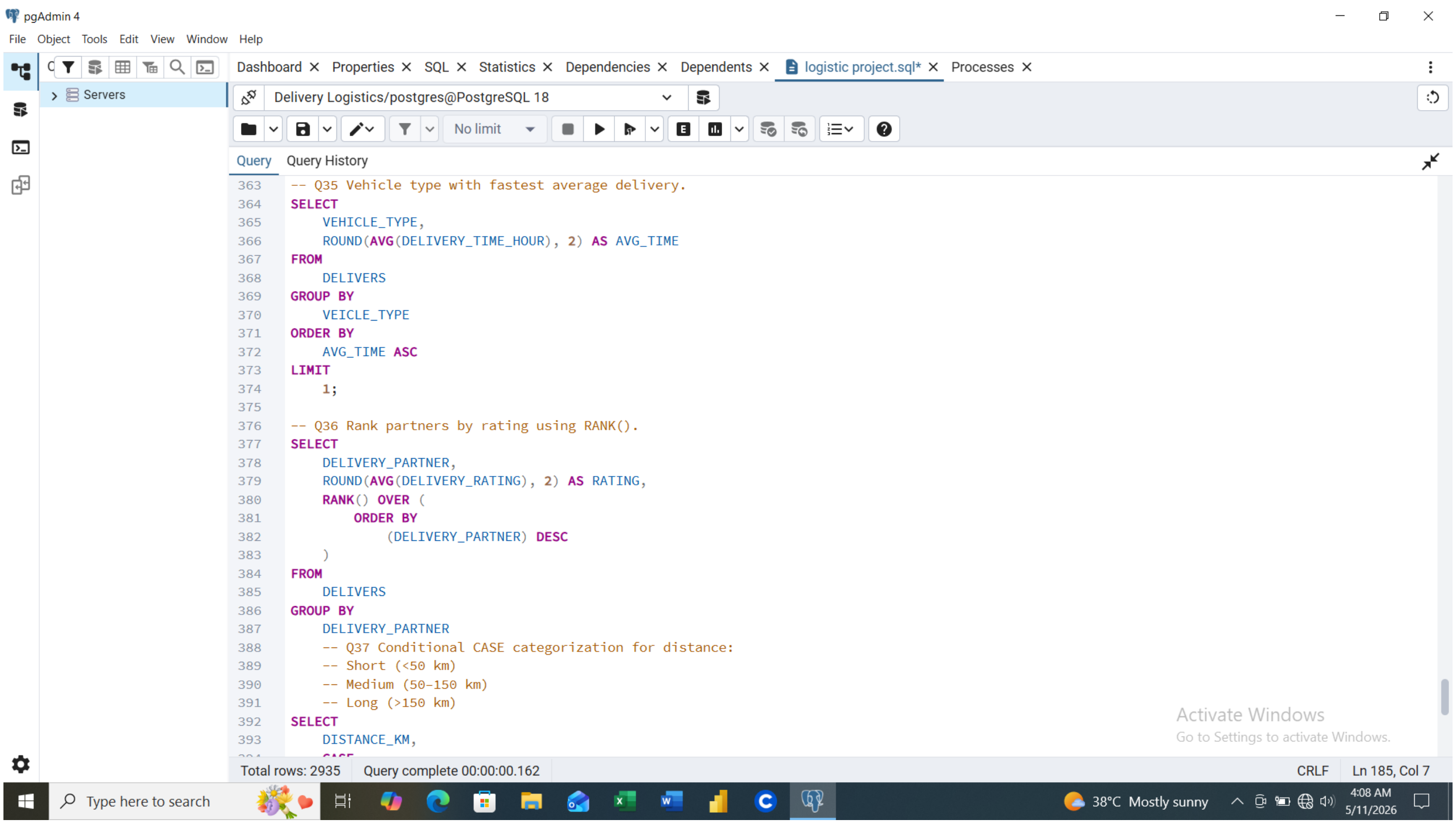


```sql
WHERE
    DELIVERY_STATUS = 'Failed'
GROUP BY
    DELIVERY_PARTNER
ORDER BY
    FAIL_DELIVERS DESC
LIMIT
    1;

-- Q31 Average rating of delayed vs non-delayed.
SELECT
    DELIVERY_STATUS,
    ROUND(AVG(DELIVERY_RATING), 2)
FROM
    DELIVERS
GROUP BY
    DELIVERY_STATUS;

-- Q32 Most profitable delivery mode (highest avg cost).
SELECT
    DELIVERY_MODE,
    ROUND(AVG(DELIVERY_COST), 2) AS AVG_COST
FROM
    DELIVERS
GROUP BY
    DELIVERY_MODE
ORDER BY
    AVG_COST DESC
LIMIT
    1;

-- Q33 Weather impact on delay percentage.
```

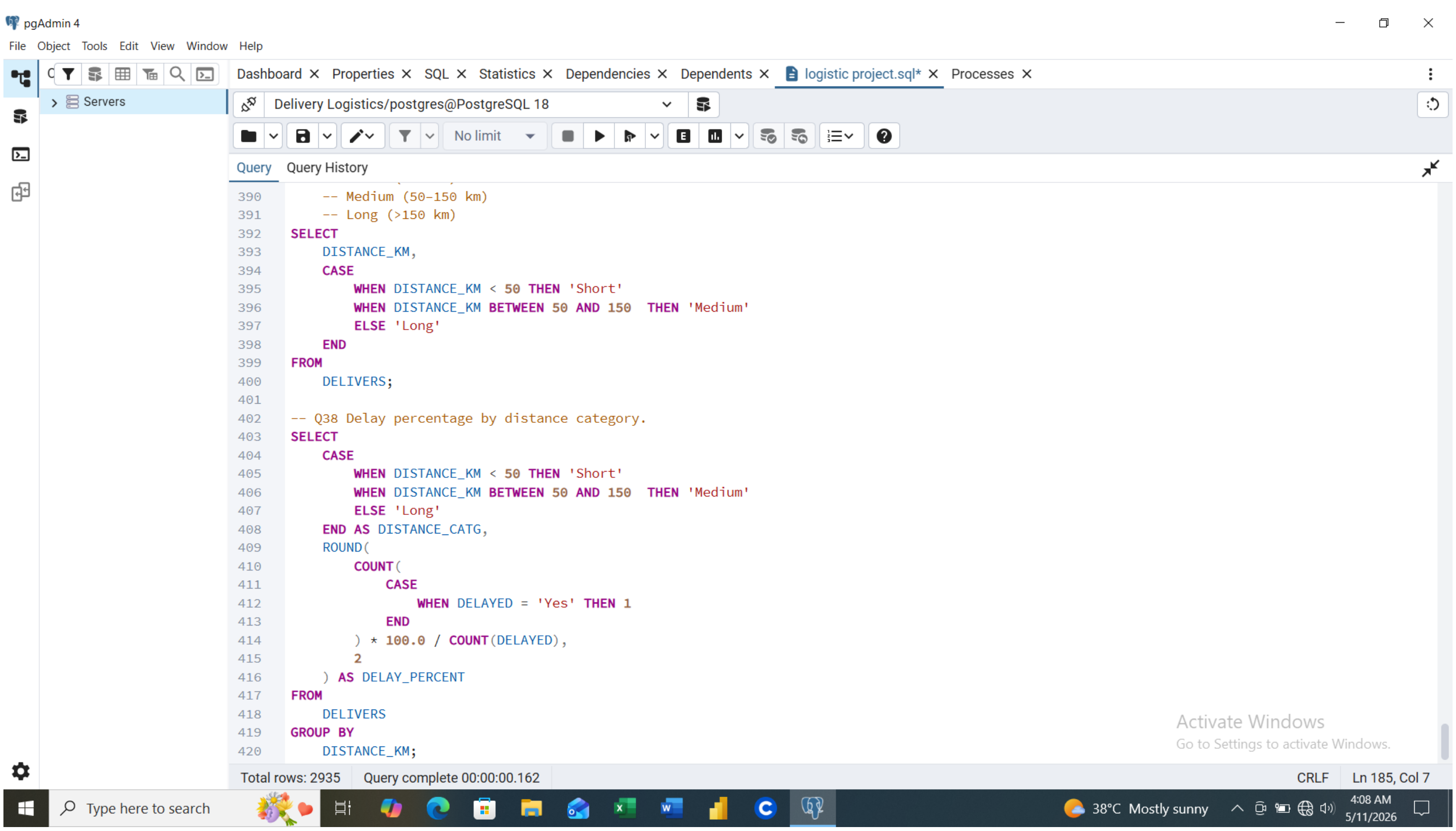
Total rows: 2935 Query complete 00:00:00.162

```sql
-- Q33 Weather impact on delay percentage.
SELECT
    WEATHER_CONDITION,
    ROUND(
        COUNT(
            CASE
                WHEN DELAYED = 'Yes' THEN 1
            END
        ) * 100.0 / COUNT(DELAYED),
        2
    )
FROM
    DELIVERS
GROUP BY
    WEATHER_CONDITION;

-- Q34 Package type with highest cost per kg.
SELECT DISTINCT
    PACKAGE_TYPE,
    ROUND(AVG(DELIVERY_COST / PACKAGE_WEIGHT_KG), 2) AS COST_PER_KG
FROM
    DELIVERS
GROUP BY
    PACKAGE_TYPE
ORDER BY
    COST_PER_KG DESC
LIMIT
    1;

-- Q35 Vehicle type with fastest average delivery.
SELECT
    VEHICLE_TYPE
```

Total rows: 2935 Query complete 00:00:00.162



```sql
-- Q35 Vehicle type with fastest average delivery.
SELECT
    VEHICLE_TYPE,
    ROUND(AVG(DELIVERY_TIME_HOUR), 2) AS AVG_TIME
FROM
    DELIVERS
GROUP BY
    VEICLE_TYPE
ORDER BY
    AVG_TIME ASC
LIMIT
    1;

-- Q36 Rank partners by rating using RANK().
SELECT
    DELIVERY_PARTNER,
    ROUND(AVG(DELIVERY_RATING), 2) AS RATING,
    RANK() OVER (
        ORDER BY
            (DELIVERY_PARTNER) DESC
    )
FROM
    DELIVERS
GROUP BY
    DELIVERY_PARTNER
    -- Q37 Conditional CASE categorization for distance:
    -- Short (<50 km)
    -- Medium (50-150 km)
    -- Long (>150 km)
SELECT
    DISTANCE_KM,
```



```sql
    -- Medium (50-150 km)
    -- Long (>150 km)
SELECT
    DISTANCE_KM,
    CASE
        WHEN DISTANCE_KM < 50 THEN 'Short'
        WHEN DISTANCE_KM BETWEEN 50 AND 150  THEN 'Medium'
        ELSE 'Long'
    END
FROM
    DELIVERS;

-- Q38 Delay percentage by distance category.
SELECT
    CASE
        WHEN DISTANCE_KM < 50 THEN 'Short'
        WHEN DISTANCE_KM BETWEEN 50 AND 150  THEN 'Medium'
        ELSE 'Long'
    END AS DISTANCE_CATG,
    ROUND(
        COUNT(
            CASE
                WHEN DELAYED = 'Yes' THEN 1
            END
        ) * 100.0 / COUNT(DELAYED),
        2
    ) AS DELAY_PERCENT
FROM
    DELIVERS
GROUP BY
    DISTANCE_KM;
```

Key Insights

1. A measurable percentage of deliveries experienced delays, highlighting opportunities to improve on-time performance.
2. Certain delivery partners had higher failure rates than others.
3. Delayed deliveries generally received lower customer ratings.
4. Weather conditions significantly impacted delivery delays.
5. Delivery costs varied across regions, package types, and delivery modes.
6. Some vehicle types consistently delivered faster than others.
7. Longer-distance deliveries showed a higher likelihood of delays.
8. Specific package types generated higher costs per kilogram.
9. Delivery mode influenced both cost and delivery speed.
10. Operational performance varied across partners and regions.

Conclusion

This project demonstrates how SQL can be used to transform raw logistics data into meaningful business insights. By analyzing delivery delays, failures, costs, and customer ratings, the project identified key factors affecting logistics performance, including weather conditions, delivery distance, vehicle type, and delivery partner performance.

The findings can help logistics companies optimize operations, reduce delays, improve customer satisfaction, and make informed, data-driven decisions. This project also showcases strong SQL and analytical skills relevant to entry-level Data Analyst roles.

Project Files Included

- logistics_delivery_analysis.sql
- logistics_delivery_data.csv
- Logistics_Delivery_Analysis_Report.pdf
- README.md
- screenshots/

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